

Complex Analysis solution

A1

Proof. We draw the cycle Γ as the following. Then

1. In region I, for $z \in \mathbb{C} \setminus \overline{D}_4(0)$, $\text{ind}_{\Gamma}(z) = 0$.
2. In region II, for $z \in D_4(0) \setminus (\overline{D}_1(-1) \cup \overline{D}_1(1))$, $\text{ind}_{\Gamma}(z) = 1$.
3. In region III, for $z \in D_1(-1)$, $\text{ind}_{\Gamma}(z) = 0$.
4. In region IV, for $z \in D_1(1)$, $\text{ind}_{\Gamma}(z) = 0$.

□

A2

Proof. We divide $D_4(0)$ into sub-regions as the following. Consider the cycle $\Gamma' = \sum_{i=1}^6 \partial R_i$. For each $1 \leq i \leq 6$, note ∂R_i lies in some open convex set on which f is holomorphic. For instance, the two open convex sets containing R_2 and R_4 are drawn. Thus by Cauchy theorem on convex sets, $\int_{\partial R_i} f(w)dw = 0$ for each i . It follows $\int_{\Gamma'} f(w)dw = \sum_{i=1}^6 \int_{\partial R_i} f(w)dw = 0$. Pairing up repeated paths of opposite orientations (i.e, the six straight lines), we see

$$\int_{\Gamma'} f(w)dw = \int_{\partial D_4(0)} f(w)dw + \int_{-\partial D_1(-1)} f(w)dw + \int_{-\partial D_1(1)} f(w)dw.$$

So $0 = \int_{\partial D_4(0)} f(w)dw - \int_{\partial D_1(-1)} f(w)dw - \int_{\partial D_1(1)} f(w)dw$.

□

4.3 #3

Proof. Since $z^3 e^{1/z}$ is analytic on the punctured plane $\mathbb{C} - \{0\}$, by Theorem 4.3.4, $z^3 e^{1/z}$ admits a unique Laurent series expansion of the form $\sum_{k=-\infty}^{\infty} c_k z^k$. For $z \neq 0$, consider an arbitrary compact neighbourhood $K \subseteq \mathbb{C} - \{0\}$. On K , we compute

$$z^3 e^{1/z} = z^3 \left(\sum_{n=0}^{\infty} \frac{z^{-n}}{n!} \right) \stackrel{!}{=} \sum_{n=0}^{\infty} \frac{z^{3-n}}{n!} = \sum_{k=-\infty}^3 \frac{z^k}{(3-k)!}.$$

Here $!$ holds because $\sum_{n=0}^{\infty} \frac{z^{-n}}{n!}$ converges absolutely on K by ratio test. In addition $\sum_{n=0}^{\infty} \frac{z^{3-n}}{n!}$ converges uniformly on K . Indeed, as compact set K is bounded, choose $R > 0$ such that $K \subseteq D_R(0)$. For each $n \in \mathbb{N}$, $|\frac{z^{3-n}}{n!}| \leq \frac{R^{3-n}}{n!}$. Since $\sum_{n=0}^{\infty} \frac{R^{3-n}}{n!}$ converges by ratio test, Weierstrass M-test implies $\sum_{n=0}^{\infty} \frac{z^{3-n}}{n!}$ converges uniformly on K . By uniqueness, $\sum_{n=0}^{\infty} \frac{z^{3-n}}{n!}$ must be the Laurent series. $\square$

4.5 #6

Proof. By assumption, there is some $r > 0$ such that $f(z) = (z - z_0)^k g(z)$ on $D_r(z_0)$ where $g(z)$ is analytic and $g(z_0) \neq 0$. Since $g(z)$ is continuous, there is some $0 < r_1 < r$ such that $|g(z)| \neq 0$ on $D_{r_1}(z_0) \setminus \{z_0\}$. This implies $|f(z)| = |z - z_0|^k |g(z)| > 0$ on $D_{r_1}(z_0) \setminus \{z_0\}$.

Let $V = D_{r_2}(z_0)$ for some $0 < r_2 < r_1$. Then $f(z) \neq 0$ on $\overline{V} \setminus \{z_0\}$. So by construction, $f(z)$ has only one zero of order k at z_0 on V . That is, $f(z)$ has exactly k roots on V .

Since $f(z)$ continuous on the compact set ∂V , there is some $M > 0$ such that $|f(z)| \geq M$ on ∂V . Let $W = \{w : 0 < |w| < M\}$. We show $f(z) - w$ and $f(z)$ have the same number of zeroes on V .

Clearly $f(z) - w$ is analytic on V . Then note $f(z) - w$ has no zeroes on ∂V because

$$|f(z) - w| \geq ||f(z)| - |w|| \geq M - |w| > 0.$$

Finally we note $|(f(z) - w) - f(z)| = |w| < M \leq |f(z)|$ on V . By Rouché's theorem, $f(z) - w$ and $f(z)$ have the same number of zeroes on V . But we have showed $f(z)$ has exactly k roots on V . $\square$